MIL-S-45666(Ord)
22 April 1960
SUPERSEDING
FA(FC)-PD-9
2 May 1958
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30 April 1958

MILITARY SPECIFICATION

SIGHTUNIT, OPTICAL: XM43

1. SCOPE

1.1 This specification covers a basic type of instrument which is known as Optical Sightunit: XM43, and is used on the "Honest John" Rocket Launcher, XM33. This sightunit is composed of the following components:

Telescope, Elbow: M62A1D
Adapter, Telescope: M9A1
Mount, Telescope: 8228899

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

O-E-760	Ethyl, Alcohol (Ethanol)
UU-P-313	Paper, Lens Tissue
UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
CCC-C-440	Cloth, Cotton Cheesecloth, Bleached and Unbleached
JJJ-C-561	Cotton; Sterile and Nonsterile
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-601	Boxes; Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-B-676	Boxes; Set-up, Paperboard
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant

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Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed Flexible (Waterproofed)
MIL-L-3150	Lubricating Oil, Preservative, Medium
MIL-C-7769	Cushioning Material, Bound
MIL-T-13321	Telescope, Elbow: M62 (Series)
MIL-A-13325	Adapter, Telescope: M9 (Series)
MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-13988	Paper, Lens Tissue, Antitarnish, Wrapping

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F8228899	Mount, Telescope: 8228899
F8244830	Adapter, Telescope: M9A1
F8267901	Sightunit Optical: XM43
F8572120	Telescope, Elbow: M62A1D

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one sightunit, representative of production, shall be made available to the Government to determine if the sightunit meets all the requirements of this specification.

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3.2 Materials.- The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The sightunit shall be manufactured in accordance with Drawing F8267901, "Sightunit, Optical: XM43" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product prior to final acceptance by the Government.

3.3.1 Components.- The components listed in 1.1 shall conform to the requirements of their respective specifications and drawings prior to assembly as a sightunit and shall be assembled in accordance with Drawing F8267901.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specifications MIL-F-13926 and MIL-O-13830.

3.4.1 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.4.2 MIL-O-13830

(a) Cleanliness

3.5 Performance.- With the dovetail of the sightunit horizontal and the bubble in the cross level vial centered, the sightunit shall meet the following requirements.

3.5.1 Horizontal travel.- With the line of sight horizontal and coincident with a horizontal target line, the line of sight shall not deviate from the target line by more than 1.7 mils (total spread 3.4 mils) when the adapter is rotated through 6400 mils by means of the adapter knob.

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3.5.2 Plumb travel.- The line of sight shall not deviate from a plumb line by more than plus or minus 0.5 mil (total deviation not to exceed 0.5 mil) through the entire range of elevation and depression.

3.5.3 Cross level vial.- When determining plumb travel accuracy, the cross level bubble shall remain central within 1/2 a graduation space on the level vial when the dovetail plate is rotated about its axis to return the line of sight to horizontal.

3.6 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

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4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.4 Production sample.- Unless otherwise specified in the contract, one sightunit shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production sample failure.- Should the production sample fail to pass the prescribed examination and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Inspection provisions.

4.5.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

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4.5.2 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the Sightunit, Optical: XM43. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.5.2.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.5.2.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.6 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.7 Acceptance tests.

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4.7.1 Classification of defects.- The classification of defects in Tables II, III & IV shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 1.0%	Req't Par.	Q.A. Par.	Test Method
1. Cleanliness	3.4.2		MIL-C-13830
2. Performance	3.5	4.9.1	Device, Special Testing (DST)
3. Horizontal travel	3.5.1	4.9.2	DST
4. Plumb travel	3.5.2	4.9.3	DST
5. Cross level vial	3.5.3	4.9.4	DST
6. Identification marking	3.6	4.9.5	Visual

4.8 Inspection equipment.

4.8.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

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4.8.1.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.9 Test methods and procedures.

4.9.1 Performance.- The sightunit shall be mounted on a special testing device capable of positioning the dovetail axis of the unit in a horizontal plane, with the cross-level vial bubble centered with respect to the vial graduations as indicated in 3.5 when the sightunit is cross-leveled.

4.9.2 Horizontal travel.- With the sightunit mounted on the special testing device, indicated above which has been leveled to traverse a horizontal plane, a target is located in a horizontal plane with the line of sight of the elbow telescope of the unit. Take four readings at 0, 1600, 3200, and 4800 mils in azimuth by rotation of the adapter knob. Deviation shall not exceed the requirement of 3.5.1.

4.9.3 Plumb travel.- Plumb travel test is accomplished by means of an elevation testing device capable of displacing the sightunit in a vertical plane to permit reading of the entire range of elevation and depression when focused on a plumb line. The device shall be provided with a mounting surface for supporting the dovetail of the sightunit in a horizontal position, and provide means for plumb travel. The telescope's vertical reticle line shall be observed with respect to the target plumb line and shall not exceed the limits specified in 3.5.2 when the line of sight traversed the full movement by means of the elevation knob of the sightunit and return to starting position by means of the test fixture.

4.9.4 Cross-level vial.- When the sightunit is returned to the level position by means of the elevation test fixture as specified in 4.9.3 above, note the position of the level bubble which shall be central within the limits specified in 3.5.3.

4.9.5 Identification marking.- The identification marking shall be visually inspected for compliance with Standard MIL-STD-130.

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4.10 Preservation and packaging tests.- The test procedures in Specification MIL-P-116 shall be used for the following packaging and preservation tests to determine the minimum acceptable quality.

4.10.1 Sampling for test.- Samples for the tests listed in Table V shall be selected from lots that have met the minimum inspection requirements listed in Table III or IV.

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

Sightunit without Canvas Container, Carrying Case and Instrument Light

MAJOR: AQL 1.0%	Req't Par.	Test Method
1. Cleanliness of sightunit and packaging materials (Levels A,C)	5.3.1.2 5.3.2.1 5.7	MIL-P-116-Visual
2. Preservative application not as specified (Level A)	5.3.1.4	MIL-L-3150-Visual
3. Preservative coated surface not wrapped as specified (Level A)	5.3.1.5.1.1	MIL-B-121-Visual
4. Level vials not protected as specified (Levels A,C)	5.3.1.5.1.2 5.3.2.2.1	Visual
5. Eyeshield not wrapped and secured to sightunit as specified (Levels A,C)	5.3.1.5.1.3 5.3.2.2.1	SME-Visual
6. Sightunit not adjusted and wrapped as specified (Levels A,C)	5.3.1.5.1.4 5.3.2.2.1	Visual
7. Water-vaporproof bag damaged, improperly sealed (Level A)	5.3.1.5.1.5	MIL-B-117-Visual
8. Sightunit not immobilized in fiberboard box (Levels A, C)	5.3.1.5.1.6 5.3.2.2.1	Visual
9. Unit packs not immobilized in shipping container (Levels A,B, C)	5.4.1 5.4.2 5.4.3	Visual
10. Excessive gross weight (Level C)	5.4.3.1	Standard Measuring Equipment (SME)
11. Markings illegible, incorrect, not conforming to specifications (Levels A,B,C)	5.5	MIL-STD-129-Visual

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MINOR: AQL 4.0%	Req't Par.	Test Method
41. Fiberboard box not conforming to specifications (Levels A,C)	5.3.1.5.1.6 5.3.2.2.1	PPP-B-636-Visual-SME
42. Shipping container not conforming to specifications (Levels A,B,C)	5.4.1 5.4.2 5.4.3	SME-Visual-PPP-B-601 PPP-B-585-PPP-B-621 PPP-B-636
43. Strapping improperly applied, missing (Levels A,B.C)	5.4.4	Visual
44. Workmanship	5.7	Visual

TABLE IV- CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

Sightunit with Canvas Container, Carrying Case and Instrument Light

MAJOR: AQL 1.5%	Req't Par.	Test Method
1. Cleanliness of sightunit, canvas container, carrying case, instrument light and packaging materials (Levels A,C)	5.3.1.2 5.3.2.1 5.7	Visual MIL-P-116
2. Preservative application not as specified (Level A)	5.3.1.4	Visual MIL-L-3150
3. Preservative coated surfaces not wrapped as specified (Level A)	5.3.1.5.2.1	Visual MIL-B-121
4. Level vials not protected as specified (Levels A,C)	5.3.1.5.2.1 5.3.2.2.2	Visual
5. Eyeshield not wrapped and secured to sightunit as specified (Levels A,C)	5.3.1.5.2.1 5.3.2.2.2	Visual SME
6. Instrument light not positioned as specified (Levels A,C)	5.3.1.5.2.2 5.3.2.2.2	Visual
7. Sightunit not placed in canvas container as specified (Levels A,C)	5.3.1.5.2.3 5.3.2.2.2	Visual
8. Carrying case w/canvas container not wrapped as specified (Levels A,C)	5.3.1.5.2.4 5.3.2.2.2	Visual
9. Water-vaporproof bag damaged, improperly sealed (Level A)	5.3.1.5.2.5	Visual MIL-B-117
10. Carrying case not immobilized in fiberboard box (Level A,C)	5.3.1.5.2.5 5.3.2.2.2	Visual SME
11. Unit packs not immobilized in shipping container (Levels A,B,C)	5.4.1 5.4.2 5.4.3	Visual

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TABLE IV- CLASSIFICATION OF DEFECTS

MAJOR: AQL 1.5% (Cont'd)	Req't Par.	Test Method
12. Excessive gross weight (Level C)	5.4.3.1	SME
13. Markings, illegible, incorrect, not conforming to specifications (Levels A,B,C)	5.5	Visual

MINOR: AQL 4.0%	Req't Par.	Test Method
41. Fiberboard box not conforming to specifications (Levels A,C)	5.3.1.5.2.5 5.3.2.2.2	SME-Visual PPP-B-636
42. Shipping container not conforming to specifications (Levels A,B,C)	5.4.1 5.4.2 5.4.3	PPP-B-601-SME PPP-B-621-PPP-B-585 PPP-B-636
43. Strapping improperly applied, missing (Levels A,B,C)	5.4.4	Visual
44. Workmanship (Levels A,B,C)	5.7	Visual

TABLE V - CLASSIFICATION OF DEFECTS

Use Inspection Level L6, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 6.5%	Req't Par.	Test Method
1. Heat seal (Level A)	5.3.1.5.1.5 5.3.1.5.2.5	MIL-P-116
2. Quick leak (Level A)	5.3.1.5.1.5 5.3.1.5.2.5	MIL-P-116

4.9.2 Preproduction pack.- The preproduction pack specified in 5.6 shall be inspected 100 percent by the contractor or his representative and witnessed by the Government inspector to determine conformance with the requirements of this specification for level "A".

4.9.2.1 Preproduction pack failure.- Should the preproduction pack fail to pass the test for Government acceptance, packaging operations will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new pack shall be completely inspected by the contractor and witnessed by the Government inspector.

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4.9.2.2 Rough handling test.- The preproduction pack specified in 5.6 shall be subjected to the rough handling test in accordance with Specification MIL-P-116 applicable only to level "A".

4.9.3 Workmanship shall comply with the requirements of 5.7.

5. PREPARATION FOR DELIVERY

5.1 The materials and methods.- The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

5.1.1 Materials.- The following shall apply wherever packaging materials are referred to in this specification by name only:

REFERENCE	FULL IDENTIFICATION
Gummed tape	Class 2 of Specification UU-T-111
Paperboard cushioning	Type I, Class I of Specification PPP-P-291
Pressure-sensitive tape	Type I or II of Specification UU-T-106

5.2 Size of container.- Bags, interior containers and exterior containers shall be of a minimum size that will adequately contain the items.

5.3 Preservation and packaging.

5.3.1 Level A

5.3.1.1 Disassembling.- The eyeshield shall be removed from the sightunit.

5.3.1.2 Cleaning

5.3.1.2.1 Optics

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5.3.1.2.1.1 The following materials and equipment shall be used in the cleaning of optics:

Materials and equipment	Specification
Alcohol, ethyl	Grade II of Specification O-E-760
Bottle, glass, 4 ounce	----------
Brush, camel's hair	----------
Cheesecloth	Specification CCC-C-440
Cloth, batiste	----------
Cotton, absorbent	Grade A, class 1 of Specification JJJ-C-561
Paper, lens tissue	Specification UU-P-313
Sticks, orange or equal	----------
syringe, rubber	----------

5.3.1.2.1.2 Cleaning exposed glass surfaces of optics shall be accomplished in the following manner:

(a) Loose particles of dust shall be removed from the surfaces of the optics with a camel's hair brush.

(b) The surface to be cleaned shall be wiped with a circular motion using cotton, lens paper or freshly laundered cheesecloth which is saturated with ethyl alcohol. The surfaces shall be dried with clean cotton, lens paper or freshly laundered cheesecloth.

(c) A swab shall be made by wrapping one piece of lens tissue paper around the end of an orange stick or equal. One or two drops of alcohol shall be added on the tip of the swab. The swabbing operation shall begin at the center of the polished surface. A light, downward pressure shall be exerted on the swab and with a circular motion gradually increase the radius of area being cleaned until the entire surface has been swabbed. This operation shall be repeated until no trace of dirt, lint or smears are preceptible.

(d) A rubber syringe shall be used as a bellows to remove any residual dust or lint that may remain on the cleaned surface.

(e) Immediately after cleaning, the optics shall be covered or wrapped with lens tissue paper conforming to Specification MIL-P-13988, which shall be secured in place with pressure-sensitive tape.

5.3.1.2.2 Other surfaces.- Cleaning Process C-1 of Specification MIL-P-116 shall be followed to clean all other surfaces of the sightunit, canvas container, carrying case and instrument light.

5.3.1.3 Drying.- Immediately after cleaning, the items shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

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5.3.1.4 Preservative application.- A light film of preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116) shall be applied to all unprotected metal surfaces of the sightunit. Preservative shall be applied by the brushing procedure specified in Specification MIL-P-116. Excess preservative shall be removed.

5.3.1.5 Unit packaging.

5.3.1.5.1 Sightunit without a canvas container, carrying case and instrument light.- Each sightunit shall be packaged as specified herein, which is in conformance with Method IA-8 of Specification MIL-P-116.

5.3.1.5.1.1 Covering preservative coated surfaces.- The preservative coated surfaces of the sightunit shall be covered or wrapped with greaseproofed barrier material conforming to type II, grade A, class 2 of Specification MIL-B-121. The greaseproofed barrier material shall be secured in place with pressure-sensitive tape as shown in figure 1.

5.3.1.5.1.2 Covering level vials.- The metal covers on the level vials shall be positioned so that the vials are covered. The covers shall then be secured in place with pressure-sensitive tape.

5.3.1.5.1.3 Eyeshield.- The eyeshield shall be dusted with talcum, and placed in a 2 by 2 by 1 inch (inside dimensions) set-up paperboard box conforming to type I or IV, any class and style of Specification PPP-B-676. The box shall be closed and secured with gummed tape. The boxed eyeshield shall then be secured to the sightunit as shown in figure 1 with pressure-sensitive tape.

5.3.1.5.1.4 Instrument setting and wrapping.- The knobs shall be adjusted until the configuration of the sightunit is as shown in figure 1, which will be the minimum cube for packaging purposes. The sightunit shall then be wrapped with at least two thicknesses of paperboard cushioning. The wrap shall be secured in place with gummed tape.

5.3.1.5.1.5 Water-vaporproof bag.- The wrapped sightunit shall be placed in a water-vaporproof bag conforming to type I, class e of Specification MIL-B-117. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specificatior MIL-P-116. Air shall be exhausted from the bag prior to final closure.

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5.3.1.5.1.6 Fiberboard box.- Each bagged sightunit shall be cushioned and immobilized (specified in 5.3.1.5.1.7) in a fiberboard box conforming to class 2 or 3, any compliance symbol and any style, except FTC, TS, OPF and FPF of Specification PPP-B-636. The inside dimensions of the box shall be 17 by 11 by 6 inches.

5.3.1.5.1.7 Cushioning, immobilizing and closure of fiberboard box.- The interior surfaces (sides, end, top and bottom) of the fiberboard box shall be lined with one inch thick bound cushioning material conforming to type IV of Specification MIL-C-7769. The bagged sightunit shall fit snugly in the bound cushioning material. The fiberboard box shall be closed with pressure-sensitive tape conforming to Specification PPP-T-76 in accordance with the appendix of the box specification.

5.3.1.5.2 Sightunit with canvas container, carrying case and instrument light.- Packaging shall be as specified herein, which is in conformance with Method IA-8 of Specification MIL-P-116.

5.3.1.5.2.1 Covering preservative coated surfaces, covering level vials and packaging the eyeshield.- Covering preservative coated surfaces, covering level vials and packaging the eyeshield shall be as specified in 5.3.1.5.1.1, 5.3.1.5.1.2 and 5.3.1.5.1.3.

5.3.1.5.2.2 Instrument light.- The instrument light shall be placed in the pocket provided on the exterior of the canvas container. The lead wires shall be placed through the hole in the flap of the instrument light pocket and coiled so as to fit in the other pocket of the canvas container. The coiled lead wires shall then be placed in the pocket and all flaps shall be secured with the button type fasteners.

5.3.1.5.2.3 Sightunit setting and placing the sightunit in its canvas container.- The knobs shall be adjusted until the configuration of the sightunit is the same as shown in figure 2. The sightunit shall then be placed in the canvas container and the draw strings pulled tight.

5.3.1.5.2.4 Carrying case and wrapping.- The canvas container containing the sightunit shall be placed in the carrying case. The lid of the carrying case shall then be closed and secured. The carrying case containing the items shall be wrapped with paperboard cushioning. The wrap shall be secured in place with gummed tape.

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5.3.1.5.2.5 Water-vaporproof bag and fiberboard box.- The wrapped carrying case shall be placed in a water-vaporproof bag conforming to the characteristics (except for size) of the bag specified in 5.3.1.5.1.5. Air shall be exhausted and the bag shall be sealed as specified in 5.3.1.5.1.5. The bagged carrying case shall then be immobilized in a fiberboard box conforming to the characteristics of the box specified in 5.3.1.5.1.6, except that the inside dimensions shall be 18 by 10 by 10 inches. The fiberboard box shall be closed as specified in 5.3.1.5.1.7.

5.3.2 Level C.

5.3.2.1 Disassembling, cleaning and drying.- Disassembling, cleaning and drying shall be as specified in 5.3.1.1, 5.3.1.2 and 5.3.1.3.

5.3.2.2 Unit packaging.

5.3.2.2.1 Sightunit without canvas container, carrying case and instrument light.- Packaging shall be as specified in 5.3.1.5.1.2 to 5.3.1.5.1.7 inclusive, except as follows, which is in conformance with Method III of Specification MIL-P-116:

(a) The water-vaporproof bag, specified in 5.3.1.5.1.5, shall be omitted.

(b) A fiberboard box, conforming to type I or II, class 1, any style except FTC, TS, OPF and FPF of Specification PPP-B-636, shall be used in lieu of the fiberboard box specified in 5.3.1.5.1.6. The inside dimensions of the box shall be 15 by 9 by 5 inches. The fiberboard box shall be closed as specified in the appendix of the fiberboard box specification.

5.3.2.2.2 Sightunit with canvas container, carrying case and instrument light.- Packaging shall be as specified in 5.3.1.5.2.1 to 5.3.1.5.2.5 inclusive, except as follows, which is in conformance with Method III of Specification MIL-P-116:

(a) Since no preservative is applied, covering preservative coated surfaces, specified in 5.3.1.5.2.1, is not applicable.

(b) The water-vaporproof bag, specified in 5.3.1.5.2.5, shall be omitted.

(c) A fiberboard box, conforming to the characteristics of the box specified in 5.3.2.2.1 (b) shall be used in lieu of the box specified in 5.3.1.5.2.5. The inside dimensions of the box shall be 18 by 10 by 10 inches. The box shall be closed as specified in 5.3.2.2.1 (b).

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5.4 Packing.

5.4.1 Level A.- Ten sightunits packaged as specified in 5.3.1.5.1 or six sightunits packaged as specified in 5.3.1.5.2 shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-601 - Style A or B (oversea type) - Type II or III, class 2 plywood
PPP-B-621 - Class 2 (oversea), style 2 or 4

5.4.2 Level B.- Ten sightunits packaged as specified in 5.3.1.5.1 or six sightunits packaged as specified in 5.3.1.5.2 shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style, except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 Class 1 (domestic), style 2 or 4

5.4.3 Level C.- A number of packaged sightunits, with or without canvas containers, carrying cases and instrument lights, shall be immobilized in a shipping cntainer conforming to one of the following specifications:

PPP-B-585 - Any style, except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), any style
PPP-B-636 - Type I or II, class 1, any style, except FTC, TS, OPF and FPF.

5.4.3.1 The gross weight for a Level C shipping container shall not exceed 200 pounds for a wood container and the gross weight for a fiberboard container shall not exceed the weight limits specified in the fiberboard box specification.

5.4.4 Strapping.- Strapping shall be applied as specified in the applicable container specification.

5.5 Marking.

5.5.1 Interior packages.- Each water-vaporproof bag and each fiberboard box shall be marked in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A unit package and omitted on the Level C unit package. Fragile marking shall be added on each fiberboard box in accordance with Standard MIL-STD-129.

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5.5.2 Shipping containers.- Each shipping container shall be marked in accordance with Standard MIL-STD-129. If the serial numbers of the sightunits in a Level A or B shipping container are in consecutive order, the serial numbers shall be indicated as specified in Standard MIL-STD-129. If the serial numbers are not in consecutive order, they shall be indicated on packing lists. Fragile markings shall be applied as specified in Standard MIL-STD-129.

5.5.3 Stock number.- The stock number (or numbers when the other items are packaged with the sightunit) to be added in the identification data shall be obtained from the contracting officer.

5.6 Preproduction pack.- Prior to quantity production, a preproduction pack, consisting of a complete and packed shipping container, shall be submitted to the Government inspector to determine conformance with the applicable requirements of this specification. A rough handling test is required.

5.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

6. NOTES

6.1 Intended use.- The sightunit is used as a indirect fire control instrument for the "Honest John" Rocket Launcher for laying the weapon in elevation and deflection (azimuth). It is removed when the rocket is fired or the launcher is transported.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

6.3 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Size and weight of packed shipping container.

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6.4.1 Sightunit without canvas container, carrying case and instrument light.- The following inside dimensions, overall cubage and gross weight are approximate for ten packaged sightunits packed in a nailed wood box for Level A:

Inside dimensions	- 23 inches long by 17-1/4 inches wide by 33-3/4 inches deep
Cubage	- 10 cubic feet
Gross weight	- 200 pounds

6.4.2 Sightunit with canvas container, carrying case and instrument light.- The following inside dimensions, overall cubage and gross weight are approximate for six packaged sightunits with canvas containers, carrying cases and instrument lights packed in a nailed wood box for Level A:

Inside dimensions	- 30-3/4 inches long by 20-1/4 inches wide by 18-1/2 inches deep
Cubage	- 8.4 cubic feet
Gross weight	- 185 pounds

6.5 Definitions.

6.5.1 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.5.3 Standard measuring equipment (SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, and so forth.

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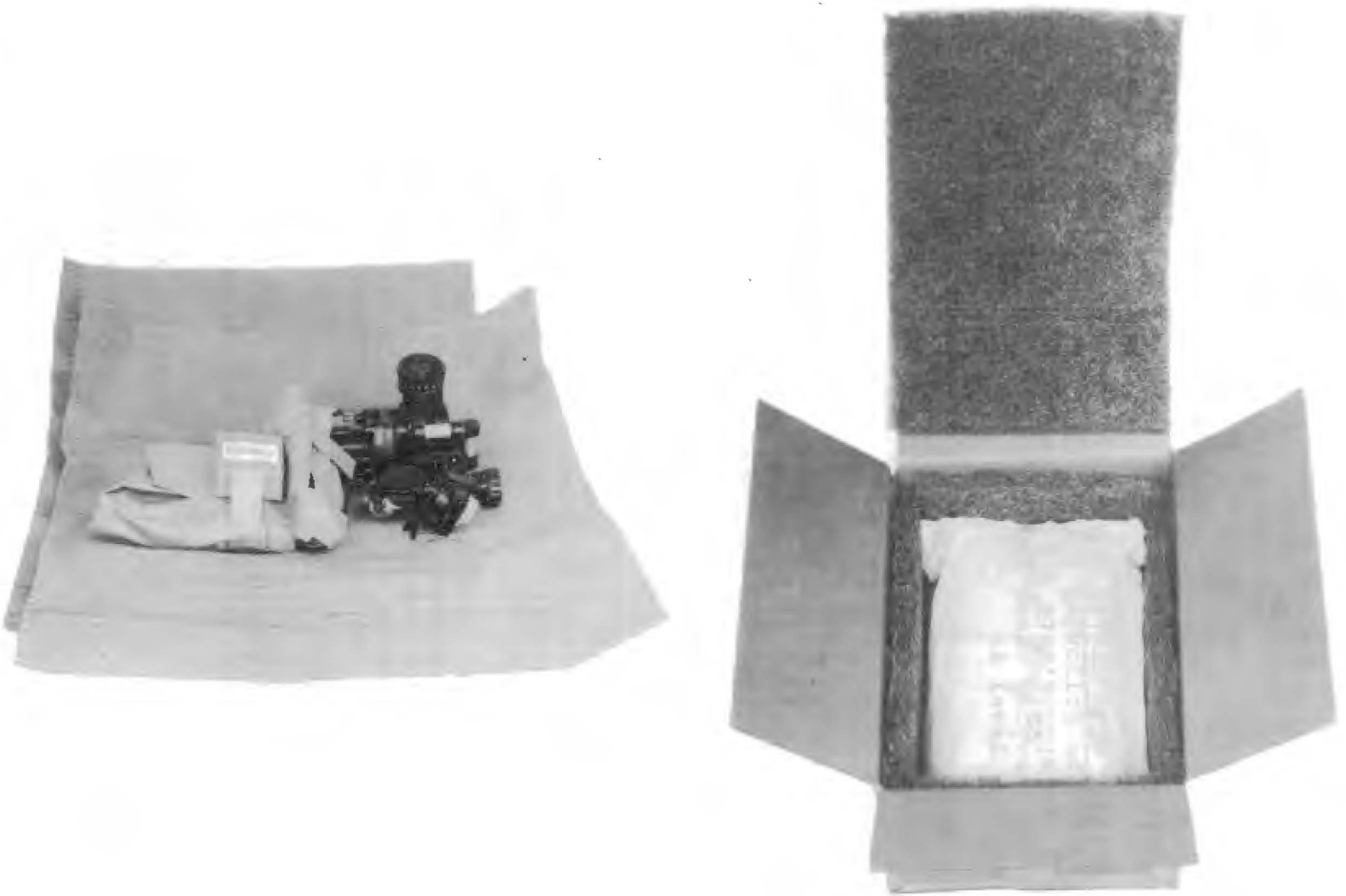
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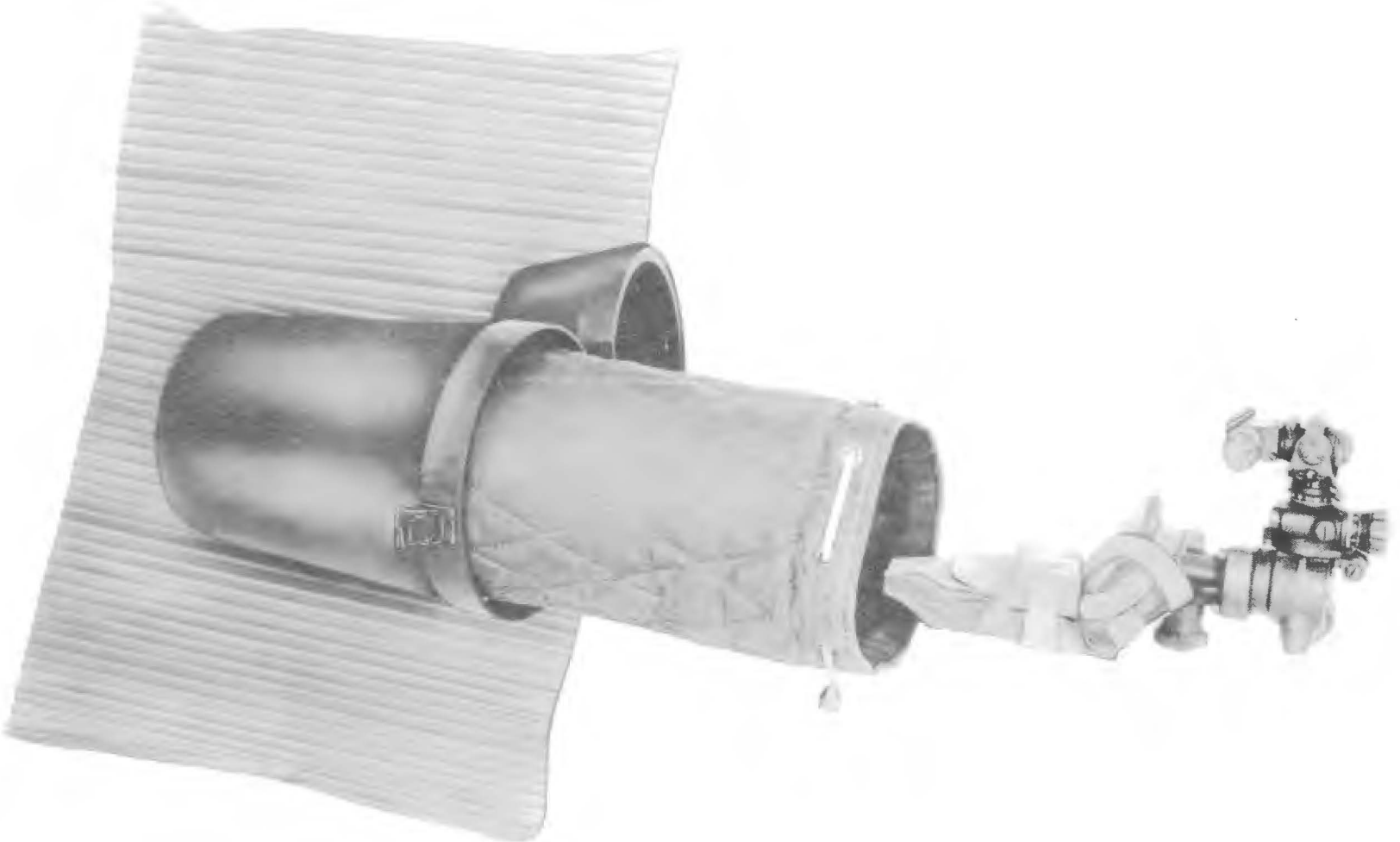
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SIGHTUNIT, OPTICAL: XM43 - Packaging the sightunit

FIGURE 1

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SIGHTUNIT, OPTICAL: XM43 - Packaging the sightunit with instrument light, canvas container and carrying case. The bagged item immobilized in a fibreboard box are not shown.

FIGURE 2

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